

Course Code : IDT 353

MPNO/MS – 24 / 1666

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

BIOLOGY FOR ENGINEERS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Draw neat and clean diagram, wherever necessary.
 - (2) Assume suitable data wherever necessary.
 - (3) All the questions are compulsory and carry marks mention against them.
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1. (a) What are the levels of organization of the human body from smallest to largest ? Give an example of each organization. 5(CO1)
(b) List the any five human physiological systems. Elaborate on the human digestive system in detail. 5(CO1)
 2. What is the necessity of different vitamins and minerals in human body.
 - (i). Classify all the vitamins found in human body into water soluble and fat soluble categories along with their advantages and disadvantages if accumulated in excess.
 - (ii). Briefly mention the function of at least 5 vitamins and Deficiency Diseases associate with them. 10(CO2)
 3. Illustrate the concept of Metabolism. Also explain the fundamental principles of Energy Transactions as applied to biology. 10(CO3)
 4. (a) What do you means by genes in biology. How many copies of genes do human have. Discuss the single gene disorders in humans along with example. 5(CO4)

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Contd.

- (b) “DNA is an information Molecule”. Describe this idea in detail. Also discuss the Concept of DNA Cloning in detail. 5(CO4)
5. Describe the various origins of bio-electric signals. Draw a typical cell potential waveform, label it properly and explain the phenomena of depolarization and repolarization. Elaborate the importance of the Na^+/K^+ ions in above process. 10(CO5)
6. (a) Explain in short :
- (i) Biosensors,
 - (ii) Bio imaging Machines. 5(CO5)
- (b) What is the ultrasound imaging system in medical imaging?
- (i) What is the frequency range of ultrasound normally used in medical field for diagnostic applications ?
 - (ii) Give any 2 medical applications of ultrasound imaging system
 - (iii) What are the any two advantages of an ultrasound imaging system ? 5(CO5)

